Secure Clinic – A Web-based Medical Record Management System

Project Description

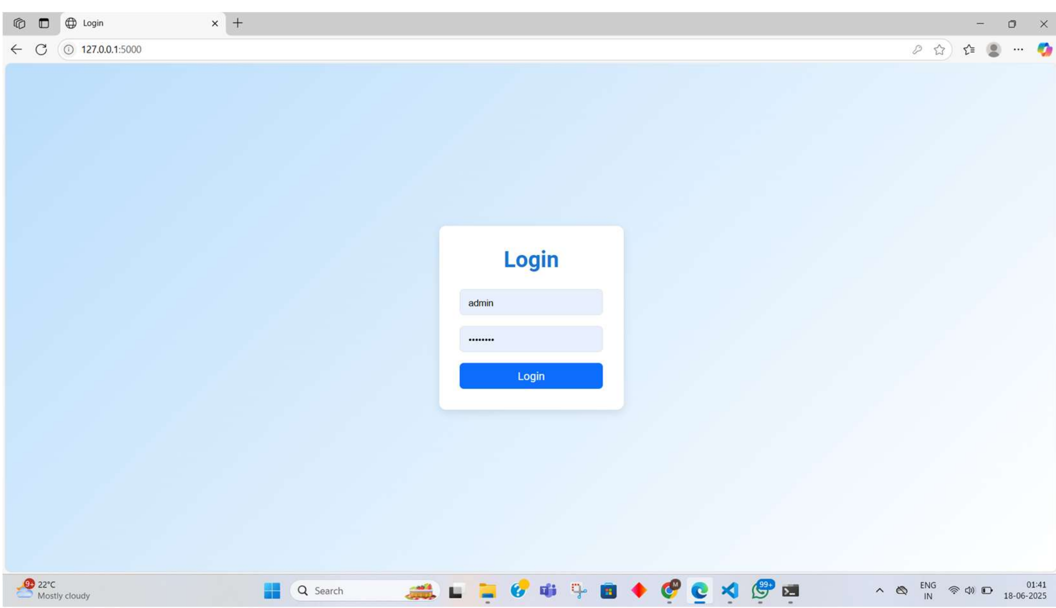
The Secure Clinic web application is a streamlined and user-friendly medical record management system developed as part of a MongoDB class project. Designed specifically for diagnostic centers and clinics, it efficiently handles patient test records using MongoDB as the primary database.

- Backend: Built with Flask (Python) for server-side logic and database interaction.
- Frontend: Clean, HTML-based interface for easy navigation by clinic staff.
- Core Features: Full CRUD operations—Create, Read, Update, Delete—for comprehensive control over medical records.
- Objective: Demonstrates proficiency in integrating MongoDB with Flask while addressing real-world challenges in organizing clinical test data.

Screenshots & Feature Highlights

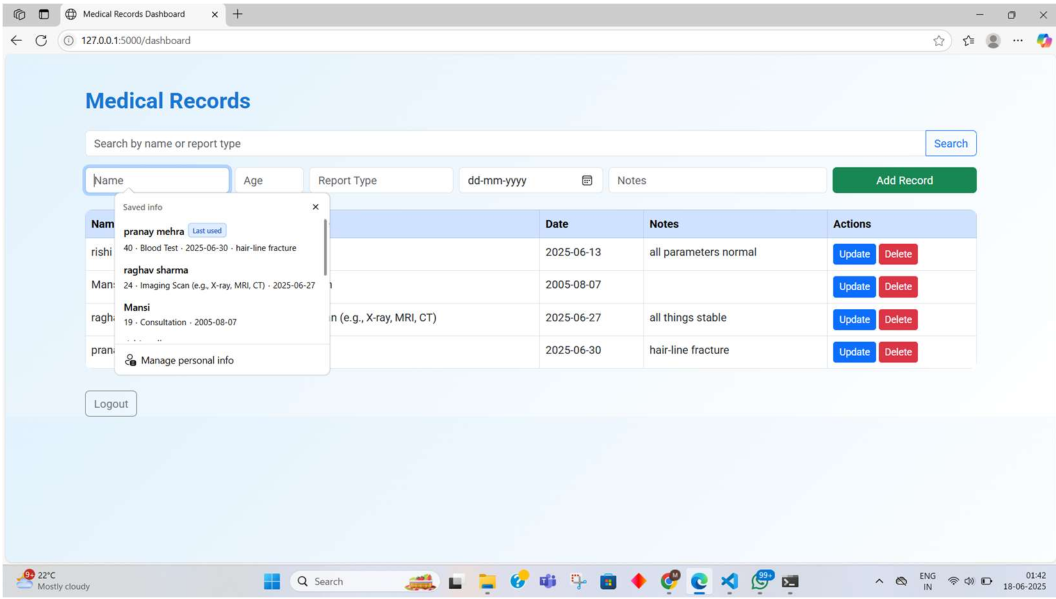
Screenshot 1: Login Page

Secure authentication interface where users enter their credentials to access the system.



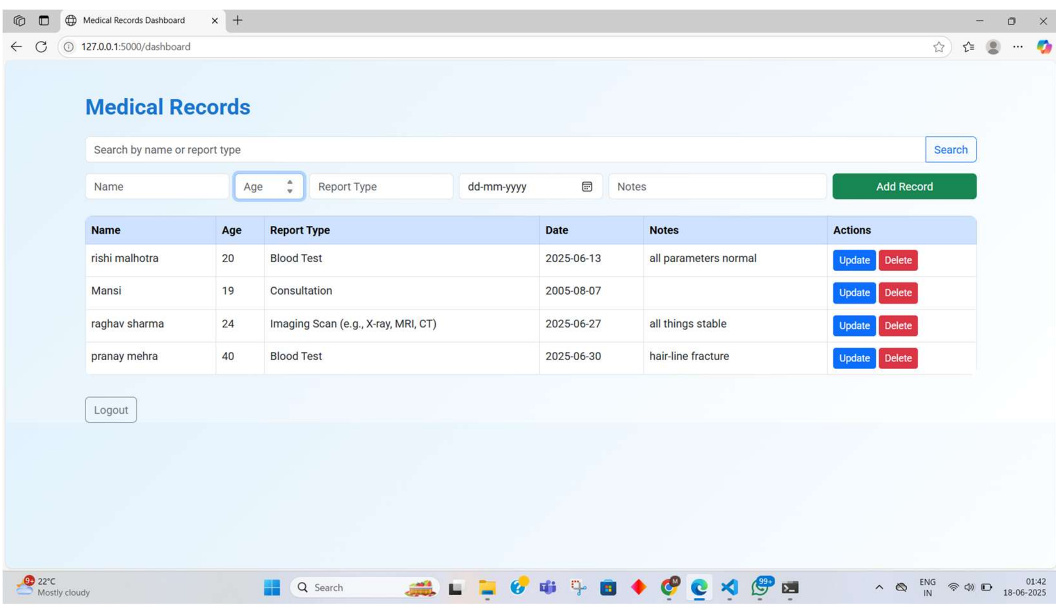
Screenshot 2: Name Entry Field

Input field allowing entry of the patient's Name—key component of the 'Create' functionality.



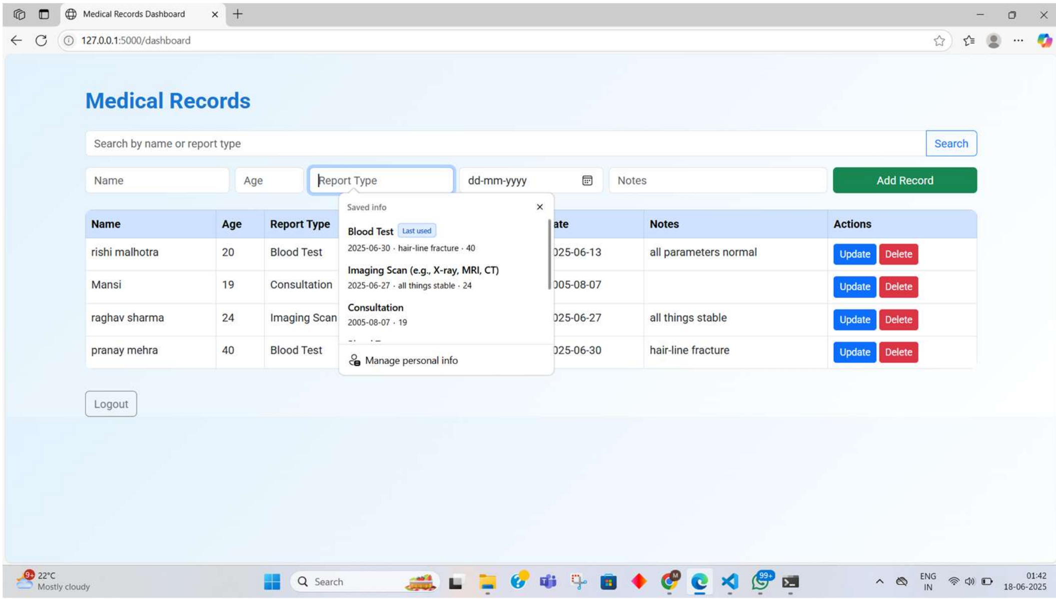
Screenshot 3: Age Entry Field

Input for the patient's Age, enabling precise patient demographic data capture.



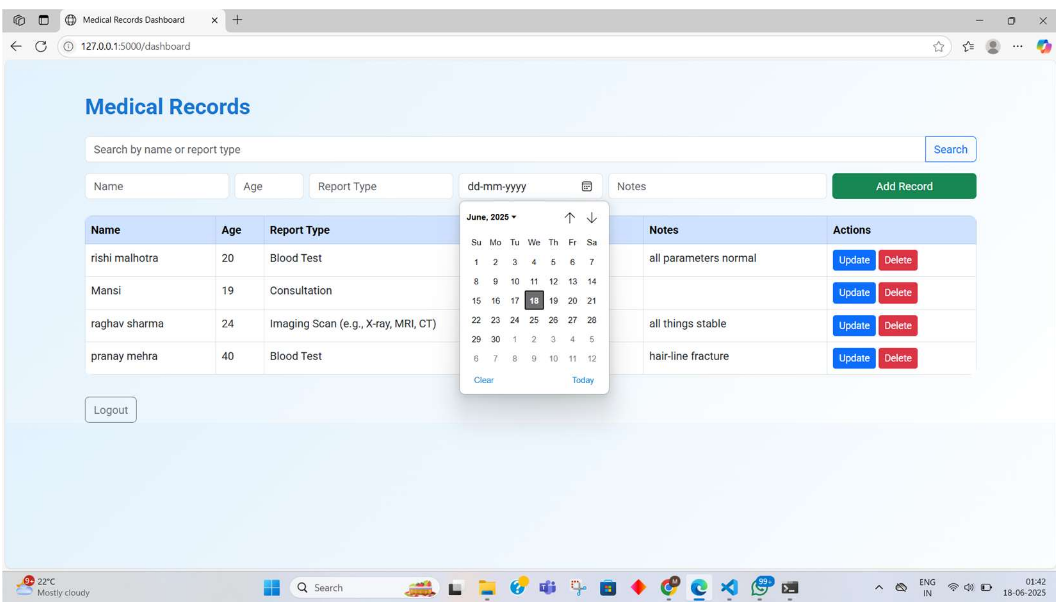
Screenshot 4: Report Type Selection

Input/dropdown for selecting the Report Type (e.g., Blood Test, Consultation, Imaging Scan).



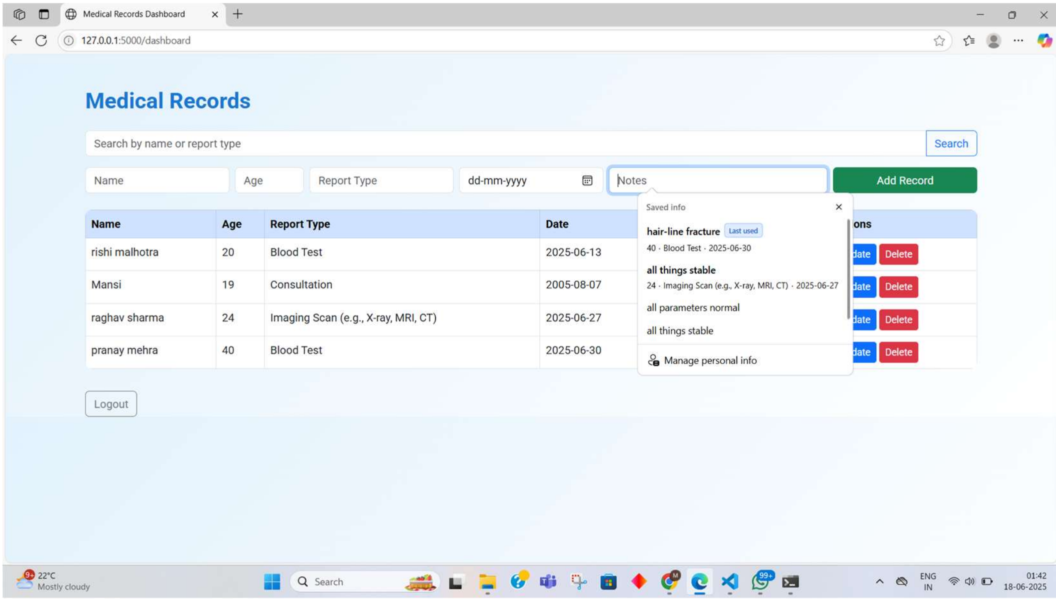
Screenshot 5: Date Picker

Calendar date-picker component for selecting the Report Date, improving data accuracy.



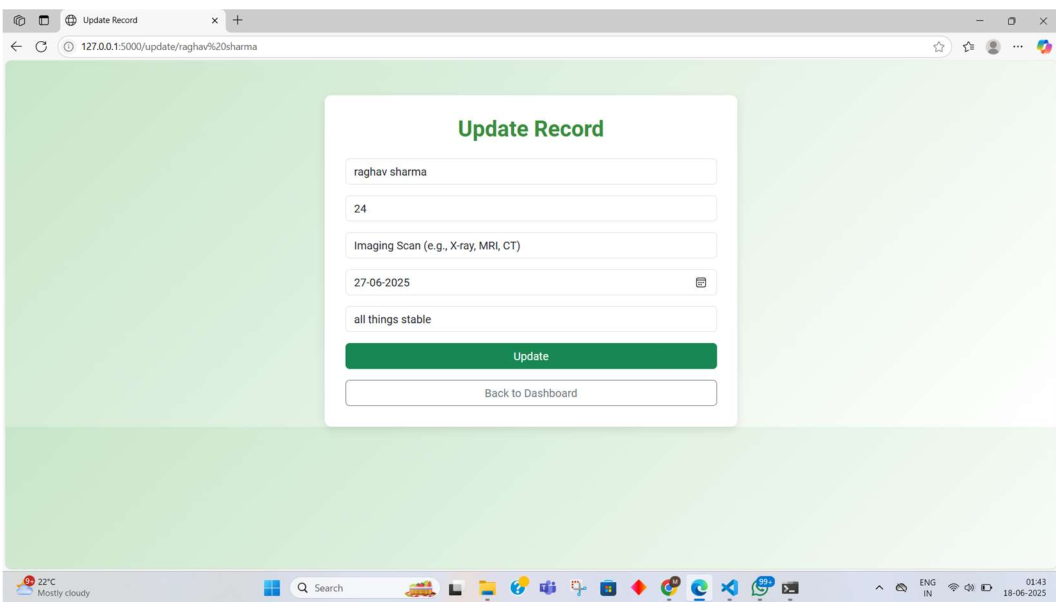
Screenshot 6: Notes Field

Free-text field for Notes, allowing clinicians to add relevant observations or comments.



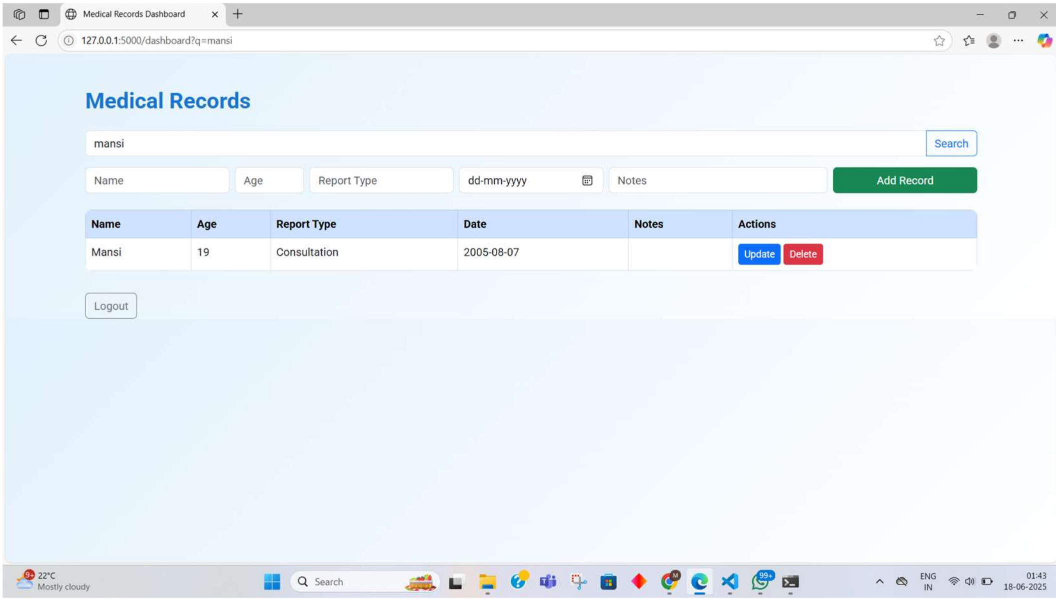
Screenshot 7: Update Button

Enables the 'Update' operation—clinicians can edit and save changes to existing records.



Screenshot 8: Delete Button

Activates the 'Delete' function to remove records securely from the database.



Technical Stack & Functionality Summary

Layer	Technology	Description
Backend	Flask (Python)	Handles HTTP requests, CRUD operations, and interacts with MongoDB
Database	MongoDB	Stores patient records with flexible schema for dynamic medical data
Frontend	HTML + CSS	Clean layout featuring input forms, tables, date-picker, and action buttons
Secure Access	Authentication	Protected login screen to ensure authorized use
Operations	CRUD	Full capability: add, view, update, delete patient records

Conclusion

Secure Clinic showcases a robust full-stack approach for managing medical records in a small clinical environment. It combines:

- Secure access controls
- Intuitive form inputs for patient data
- Flexible MongoDB storage suited for unstructured records
- Complete CRUD functionality, ensuring data integrity

This project not only demonstrates technical proficiency in Flask and MongoDB integration, but also reflects a clear understanding of practical healthcare data requirements.